

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: CSA51

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO5: *Develop the network security strategies based on the study of viruses, worms, firewall architectures, and IDS/IPS functionalities.CO (BL5 , BL6)*

Assessment Tool Weightage: 20%

QUESTIONS: 5

TOTAL MARKS: 25

Annexure A

PSEUDOCODE WRITING TASK (ALGORITHM DESIGN)

Code of Conduct:

I Eddu Suchier/192524070 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: E. Suchier

Annexure A

PSEUDOCODE WRITING TASK (ALGORITHM DESIGN)

1. Write pseudocode to simulate a simplified TLS handshake process, including client request, server authentication using a digital certificate, session key generation, and secure data exchange.

Pseudocode: Simplified TLS Handshake

Algorithm: Simplified TLS Handshake and Secure Data Exchange

Input: Client, Server, Server Digital Certificate

Output: Secure communication between Client and Server

BEGIN

1. Client sends connection request to Server
 - Client → Server: ClientHello
 - Include supported TLS version, cipher suites, and random value
2. Server responds to Client
 - Server → Client: ServerHello
 - Select TLS version and cipher suite
 - Generate and send server random value
3. Server sends Digital Certificate
 - Server → Client: Digital Certificate
4. Client verifies the certificate
 - Check certificate validity
 - Check certificate expiration
 - Verify Certificate Authority (CA) signature
 - Verify that the certificate belongs to the Server
5. IF certificate verification fails THEN
 - Terminate the connection
 - Display "Server authentication failed"
 - STOP
- END IF
6. Generate session key
 - Client and Server perform a secure key exchange
 - Derive a shared session key

7. Server and Client confirm the handshake
 - Client → Server: Finished message
 - Server → Client: Finished message
 8. Begin secure data exchange
 - Client encrypts data using the session key
 - Client → Server: Encrypted data
 - Server decrypts data using the session key
 - Server processes the data
 - Server encrypts response using the session key
 - Server → Client: Encrypted response
 - Client decrypts the response
 9. Continue encrypted communication until connection ends
 10. Terminate TLS session securely
- END

Simplified Flow

ClientHello → ServerHello → Digital Certificate → Certificate Verification → Session Key Generation → Finished Messages → Encrypted Data Exchange → Secure Termination

5. Create pseudocode for an Intrusion Prevention System (IPS) that monitors network traffic in real time and automatically blocks malicious requests while logging intrusion attempts.

Pseudocode: Intrusion Prevention System (IPS)

Algorithm: Real-Time Intrusion Detection and Prevention

Input: Network packets/requests

Output: Allowed or blocked requests, intrusion logs

BEGIN

1. Initialize IPS

- Load malicious IP addresses

- Load attack signatures

Load security rules

Start intrusion log

2. Continuously monitor network traffic

3. FOR each incoming network request DO

Capture request

Extract:

Source IP

Destination IP

Source Port

Destination Port

Protocol

Payload

4. Check the request against security rules

Check source IP

Check attack signatures

Analyze payload for malicious patterns

Check for abnormal traffic behavior

5. IF request is identified as MALICIOUS THEN

BLOCK the request

Record intrusion in log:

Timestamp

Source IP

Destination IP

Attack type

Action = "BLOCKED"

Generate security alert

ELSE

ALLOW the request

Record normal traffic if required

END IF

6. Continue monitoring network traffic

7. When IPS is stopped:

Save all logs

Close monitoring service

END

Simplified Flow

Monitor Traffic → Capture Request → Analyze Request → Check Security Rules →

- Malicious → Block → Log → Alert
- Normal → Allow → Continue Monitoring

This IPS continuously analyzes network traffic and automatically blocks malicious requests while maintaining logs of intrusion attempts for security analysis and future prevention.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient